

BURLINGTON INTL, VT, USAWMO: **726170**Lat: **44.468N**Lon: **73.150W**Elev: **330**StdP: **14.52**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **14742****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-1.7	-16.5	2.3	-5.8	-11.6	3.0	0.0	26.4	33.0	24.4	28.5	6.2	80	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	88.5	71.3	85.7	69.9	83.0	68.6	74.4	83.9	72.7	81.5	71.1	79.5	9.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.3	117.1	79.0	69.7	110.5	77.6	68.1	104.4	76.1	38.1	84.0	36.5	80.9	35.1	79.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.6	19.7	17.9	-13.9	93.8	6.2	3.3	-18.4	96.1	-22.0	98.0	-25.5	99.9	-30.1	102.3	
(5)				-14.4	77.5	6.1	1.9	-18.8	78.9	-22.4	80.0	-25.8	81.1	-30.2	82.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	20.0	22.5	31.5	44.7	57.6	66.7	71.5	69.7	62.2	49.9	38.5	27.5
	DBStd	20.09	13.05	11.26	10.81	8.82	7.93	6.74	5.42	5.59	7.51	8.32	9.20	10.24
	HDD50	3676	930	769	581	207	19	0	0	0	3	109	360	698
	HDD65	7145	1394	1189	1039	612	257	58	8	19	140	472	795	1162
	CDD50	2580	1	0	7	48	254	500	667	612	370	105	15	1
	CDD65	573	0	0	0	3	26	109	210	166	56	3	0	0
	CDH74	4714	0	0	6	42	269	946	1753	1248	430	20	0	0
	CDH80	1333	0	0	0	7	67	284	536	330	109	0	0	0
Wind	WSAvg	7.8	8.8	8.2	8.5	8.6	7.7	7.0	6.5	6.4	6.9	7.6	8.5	8.4
Precipitation	PrecAvg	34.7	1.9	1.6	2.2	2.8	3.2	3.5	3.8	3.9	3.5	3.0	3.0	2.3
	PrecMax	50.4	5.2	5.4	4.1	6.5	6.3	8.7	9.3	7.9	10.3	6.3	6.9	5.9
	PrecMin	23.3	0.4	0.2	0.4	0.7	0.3	0.8	0.8	1.2	1.4	0.5	0.9	0.4
	PrecStd	5.8	1.2	0.9	0.8	1.2	1.4	1.6	1.6	1.5	1.7	1.4	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	54.0	69.1	79.6	87.1	91.9	93.2	90.9	88.6	77.0	66.6	56.0
		MCWB	53.0	46.9	56.4	61.3	67.2	72.7	74.8	72.8	71.5	63.7	58.1	50.4
	2%	DB	47.6	46.7	57.9	71.8	80.8	86.9	88.5	86.6	82.9	71.7	61.5	50.2
		MCWB	43.7	40.6	47.9	55.9	63.4	69.6	71.9	70.7	68.6	61.2	54.3	45.7
	5%	DB	41.8	42.6	51.6	65.6	76.4	83.1	85.7	83.6	79.0	67.4	56.9	45.9
		MCWB	38.1	37.5	43.7	52.0	61.0	67.9	70.6	69.2	67.2	58.3	50.0	41.6
	10%	DB	37.5	38.6	46.9	59.8	72.3	79.4	82.7	80.9	75.0	63.1	52.8	41.4
		MCWB	33.9	34.5	40.5	48.5	59.4	66.1	69.1	68.2	65.1	55.8	46.8	37.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	48.3	58.0	64.0	70.1	75.5	77.4	75.3	73.8	67.0	60.0	51.5
		MCDB	56.7	52.4	66.7	76.1	81.4	86.0	88.6	85.4	83.4	72.6	64.7	54.8
	2%	WB	44.1	41.7	49.9	58.2	67.0	72.9	75.1	73.2	71.0	63.1	55.4	46.4
		MCDB	46.8	46.2	56.3	68.6	76.8	82.5	84.5	82.4	79.3	70.1	61.1	49.7
	5%	WB	38.5	37.7	44.6	53.6	64.2	70.6	73.1	71.5	68.8	59.3	51.2	42.1
		MCDB	41.5	41.8	50.4	62.6	73.0	79.1	81.3	79.8	76.4	65.7	55.7	45.3
	10%	WB	34.2	34.8	40.8	50.0	61.4	68.4	71.3	70.1	66.5	56.5	47.1	37.7
		MCDB	37.2	38.3	46.7	58.8	70.0	76.0	79.8	77.7	73.7	62.1	52.0	41.0
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.1	16.4	18.6	19.9	19.1	18.8	19.0	18.9	16.0	13.5	12.8
		MCDBR	16.6	17.7	23.6	28.6	26.4	24.1	21.8	21.5	23.1	21.0	19.7	15.3
		MCWBR	15.1	14.3	16.0	16.0	12.9	10.6	9.3	9.3	11.1	12.4	15.2	14.0
	5% WB	MCDBR	16.2	17.0	21.6	24.3	21.4	19.7	18.2	18.0	18.9	19.1	17.9	15.2
		MCWBR	15.3	14.3	16.1	15.7	12.3	10.3	9.2	9.1	10.2	12.9	15.4	14.6
Clear-Sky Solar Irradiance	taub	0.286	0.289	0.329	0.352	0.380	0.434	0.443	0.418	0.370	0.336	0.310	0.280	
	taud	2.346	2.339	2.371	2.405	2.380	2.231	2.214	2.277	2.399	2.499	2.496	2.425	
	Ebn at Noon	264	286	288	289	283	266	262	265	270	267	256	257	
	Edh at Noon	25	30	33	35	37	43	44	40	32	26	22	21	
All-Sky Solar Radiation	RadAvg	453	719	1075	1359	1588	1716	1754	1577	1275	747	472	334	
	RadStd	51	70	75	166	176	152	134	75	100	80	57	28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-250	+196	+97
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	N/A

Nomenclature: See separate page